

# EC-313 Digital Signal Processing

## Complex Engineering Problem — End Term Project

(CLO4 → PLO5)

### Problem Title:

Design and Implementation of a Speech Enhancement Filter for Airport Noise Suppression Using NOIZEUS Corpus in MATLAB

### Problem Statement

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Speech communication systems operating in high-noise public environments particularly airports suffer significant degradation in intelligibility and perceptual quality. Airport noise is non-stationary in character, comprising aircraft engine rumble, public address announcements, crowd babble, and impulsive transient events, making it one of the most demanding real-world noise conditions for classical signal processing. Students are required to design and implement **one** speech enhancement algorithm of their own choice in MATLAB, apply it to the **NOIZEUS** noisy speech corpus (airport noise, 5 dB SNR), and demonstrate through quantitative metrics and critical analysis that their chosen design achieves a meaningful improvement in speech quality. The choice of filter type, design method, and parameter values is deliberately left open: students may draw on any technique covered in this course or may select any available in the research work including FIR filters, IIR filters, notch filters, bandpass filters, etc. or combinations thereof, designed using MATLAB's FDA Tool or Signal Processing Toolbox functions. The engineering challenge lies not in being told what to implement, but in **justifying why** a particular design is the most appropriate for this noise type and SNR condition.

### Dataset

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Students will use the **NOIZEUS Noisy Speech Corpus**, developed by the Speech Processing Lab at the University of Texas at Dallas. The corpus contains 30 phonetically balanced IEEE sentences produced by three male and three female speakers, each corrupted by real-world noise types drawn from the AURORA database. For this project, exclusively the **airport noise condition at 5 dB SNR** is to be used. A minimum of 10 sentences must be processed and evaluated; processing all 30 is encouraged.

**Dataset access:** <https://ecs.utdallas.edu/loizou/speech/noizeus/>

**Specification:** 8 kHz sampling rate, .wav format, non-stationary airport background noise, input SNR = 5 dB

**Mandatory dataset citation:** Hu, Y. and Loizou, P. (2007). Subjective evaluation and comparison of speech enhancement algorithms. *Speech Communication*, vol. 49, pp. 588–601.

### Technical Requirements

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#### 1. Signal Analysis and Characterization

Before any filter is applied, students must perform a thorough time-frequency characterization of both the clean and noisy speech signals. Time-domain waveforms, power spectral density (PSD) plots, and spectrograms must be generated and compared in MATLAB. The input Segmental SNR (SegSNR) and PESQ score of the unprocessed noisy signal must be computed and recorded as the baseline reference. The spectral character of the airport noise must be described identifying dominant frequency bands, time-varying components, and any tonal elements and this analysis must directly motivate the filter design chosen in the next task.

#### 2. Filter Design and Implementation

Students must design and implement **one** speech enhancement design in MATLAB. The filter type and design methodology are the student's own engineering decision. Acceptable approaches include, but are not limited to: a bandpass FIR filter retaining the speech-dominant frequency range (300–3400 Hz), a high-order IIR filter designed

using Butterworth, Chebyshev, or elliptic approximations, a multi-band or composite filter combining a highpass stage with a lowpass stage, or a notch filter bank targeting identified tonal noise components. Students may use MATLAB's **FDA Tool** for interactive design and visualization, or toolbox functions such as **fir1()**, **butter()**, **cheby1()**, **ellip()**, or **designfilt()** for programmatic design.

Regardless of the approach chosen, the following must be documented and justified in the report: the filter type and design method selected; all design parameters (preprocessing, filter order, cutoff frequency or frequencies, ripple, stopband attenuation); the rationale connecting the spectral analysis of airport noise to these parameter choices; the frequency response of the designed filter (magnitude and phase); and the filter's group delay and its implications for speech intelligibility.

### 3. Quantitative Performance Evaluation

The following objective metrics must be computed for the designed filter and presented in a single results table comparing the noisy baseline against the enhanced output:

- Segmental SNR improvement (dB)
- PESQ score (scale 1 to 4.5, using MATLAB `pesq()` or the ITU P.862 binary)
- Mean Squared Error (MSE) between enhanced output and clean reference signal
- Processing time per utterance (seconds)

Results must be compared against the noisy baseline (no filter applied). Where possible, students should relate their PESQ and SegSNR values to published results in reference papers to contextualize the performance of their design.

### 4. Spectrogram-Based Qualitative Analysis

Annotated spectrograms of the clean speech, the noisy speech, and the filter-enhanced output must be provided and discussed. The analysis must identify residual noise remaining after filtering, any speech distortion introduced (particularly in the 300–3400 Hz speech band), suppression of airport noise in non-speech frames, and any artefacts such as ringing introduced by the filter's impulse response. An informal listening comparison must supplement the spectrogram observations.

### 5. Critical Analysis and Justification

Students must critically evaluate the performance of their chosen filter against the stated objective. The discussion must address: whether the SegSNR and PESQ improvements are practically meaningful at this SNR level (5 dB); where the filter succeeds and where it falls short, with specific reference to the spectrogram; why this filter type or combination was chosen over alternatives available in MATLAB; and one concrete, technically feasible modification that could improve performance (if possible i.e. in terms of complexity/performance), supported by at least one journal or conference paper published between 2022 and 2024.

## Reference Papers

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The following papers must be cited in IEEE format. Students are expected to engage with their findings rather than merely list them, using published metric values to contextualise their own results.

- [1] Y. Hu and P. Loizou, "Subjective evaluation and comparison of speech enhancement algorithms," *Speech Communication*, vol. 49, pp. 588–601, 2007. [NOIZEUS corpus paper — mandatory dataset citation]
- [2] V. Pandey et al., "Single-channel speech enhancement using implicit Wiener filter for high-quality speech communication," *International Journal of Speech Technology*, Springer, 2022. [Compares classical filter-based methods on NOIZEUS airport and station noise; provides PESQ and SegSNR benchmarks students can reference]
- [3] N. Upadhyay et al., "Psychoacoustic model-driven spectral subtraction for monaural speech enhancement," *International Journal of Speech Technology*, Springer, 2023. [Discusses noise estimation and filter parameter tradeoffs evaluated on NOIZEUS; useful background for understanding airport noise behaviour]

## Tools and Technologies

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**MATLAB** is the mandatory implementation platform. Students are expected to use the **Signal Processing Toolbox** and may optionally use the **Audio Toolbox** for PESQ scoring. The **FDA Tool** (Filter Design and Analysis Tool, launched via **fdatool** or **filterDesigner** in newer MATLAB versions) may be used for interactive filter design, visualization of frequency response, and order selection. No deep learning libraries or pre-trained models are permitted.

## Submission Requirements

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Each group must submit three separate components. Incomplete submissions will be penalized proportionally across the relevant rubric categories.

### 1. Code Files

All MATLAB source files (**.m files**) must be submitted as a separate compressed folder (**.zip**), independently of the report. If Python scripts are used for any pre- or post-processing step, those **.py files** must also be included in the same folder. Each script must carry a brief header comment stating its purpose, inputs, and outputs. Code embedded inside the report word file will not be accepted as a substitute for actual source files. The main enhancement script must be clearly named (e.g., **airport\_filter.m**) and must run end-to-end from file loading through metric computation without manual intervention.

### 2. Figure Files

All MATLAB figures must be saved and submitted as separate figure files (**.fig** format), alongside exported image copies (**.png** or **.pdf**) in a dedicated figures folder, separate from the report. Required figures include: time-domain waveform comparison (clean, noisy, enhanced), power spectral density overlay, and annotated spectrograms for the noisy and enhanced signals. Figures embedded in the report alone, without the corresponding **.fig** source files, will not receive full marks under the Implementation criterion.

### 3. Written Report

The report must be submitted in IEEE double-column format (**.doc/.docx**) and must include, in order: abstract, introduction and noise characterisation, filter design rationale and parameter justification, results (metrics table and spectrograms), critical discussion, conclusion, and references. The report must contain a **system flowchart** showing the complete processing pipeline from audio loading through filtering to metric computation, labelled with MATLAB function names or algorithm steps at each stage. Vague or decorative flowcharts will receive no credit. Results must be presented in a single concise summary table containing SegSNR (dB), PESQ score, MSE, and processing time for both the noisy baseline and the enhanced output, with no padding text around it.

## Evaluation Rubric

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| Trait                                      | Unsatisfactory<br>(0–3)                                                          | Satisfactory (4–6)                                                                | Good (7–8)                                                                                              | Exceptional (9–10)                                                                                                                                  |
|--------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Understanding of Problem 25%</b>        |                                                                                  |                                                                                   |                                                                                                         |                                                                                                                                                     |
| <b>Depth of Knowledge<br/>Weight = 15%</b> | No understanding of airport noise characteristics or filter design theory shown. | Filter is implemented but design choices are not connected to the noise analysis. | Filter type and parameters are clearly motivated by the spectral analysis of airport noise at 5 dB SNR. | Exceptional understanding of the noise-filter interaction; design decisions are argued with precision and supported by frequency response evidence. |
| <b>Literature Review<br/>Weight = 10%</b>  | No reference papers cited.                                                       | Dataset paper cited only; no engagement with findings.                            | Two papers cited with relevant discussion of their results in                                           | All three papers cited; published PESQ and SegSNR values used to contextualise                                                                      |

|                                           |                                                         |                                                                                 |                                                                                                  |                                                                                                                                                                            |
|-------------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           |                                                         |                                                                                 | relation to own design.                                                                          | own results and motivate the proposed improvement.                                                                                                                         |
| <b>Presentation and Demonstration 25%</b> |                                                         |                                                                                 |                                                                                                  |                                                                                                                                                                            |
| <b>Demonstration Weight = 25%</b>         | Results cannot be reproduced; cross-questioning fails.  | Code runs but student cannot explain design decisions under questioning.        | Clear demonstration of the filter; student explains parameter choices and metric outcomes.       | Full command of the design; articulate and technically precise defence of all decisions including why alternatives were not chosen.                                        |
| <b>Coding and Documentation 50%</b>       |                                                         |                                                                                 |                                                                                                  |                                                                                                                                                                            |
| <b>Implementation Weight = 30%</b>        | Filter produces no output or clearly incorrect results. | Filter runs but metric computation is missing or incorrect; figures are absent. | Filter implemented correctly; SegSNR, PESQ and MSE computed and tabulated; .fig files submitted. | Fully working implementation with correct metrics, annotated spectrograms, .fig files, and a flowchart accurately reflecting the code structure.                           |
| <b>Code Readability Weight = 10%</b>      | Code is disorganized and unreadable.                    | Code runs but has no comments or structure.                                     | Code is commented and reasonably structured with a clear main script.                            | Code is modular, clearly commented throughout, with descriptive variable names and a self-contained main script.                                                           |
| <b>Report Weight = 10%</b>                | No structured report submitted.                         | Report describes the filter only; no analysis or conclusion.                    | Report covers design rationale, results table, spectrogram, and flowchart.                       | Well-structured IEEE-format report with full noise characterization, justified design, crisp results table, annotated spectrogram, and critical discussion of limitations. |

## CEP Attributes

| Attribute  | WA Mapping         | Justification                                                                                                                                                                                                                                                  |
|------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WP1</b> | WK4, WK5, WK6, WK8 | Requires specialist knowledge of digital filter design, frequency-domain analysis, and perceptual speech quality metrics, applied to an engineering design task with no prescribed solution.                                                                   |
| <b>WP3</b> | WK8                | Students must independently select a filter design strategy by abstracting from spectral evidence and published literature. Different groups will arrive at different justified designs; this is intentional and reflects genuine engineering decision-making. |
| <b>WP7</b> | WK3, WK4           | The project integrates time-frequency signal analysis, filter design theory, perceptual evaluation, and technical report writing requiring students to synthesize knowledge from across the DSP course into a single coherent engineering solution.            |

**Note:** This is an open-ended problem. Students are not told which filter/filters to use that decision is the engineering challenge. Different groups are expected to choose different designs and arrive at different conclusions, provided their choices are grounded in spectral analysis, supported by quantitative results, and argued with engineering rigor. There is no single correct answer; this is what defines a Complex Engineering Problem.